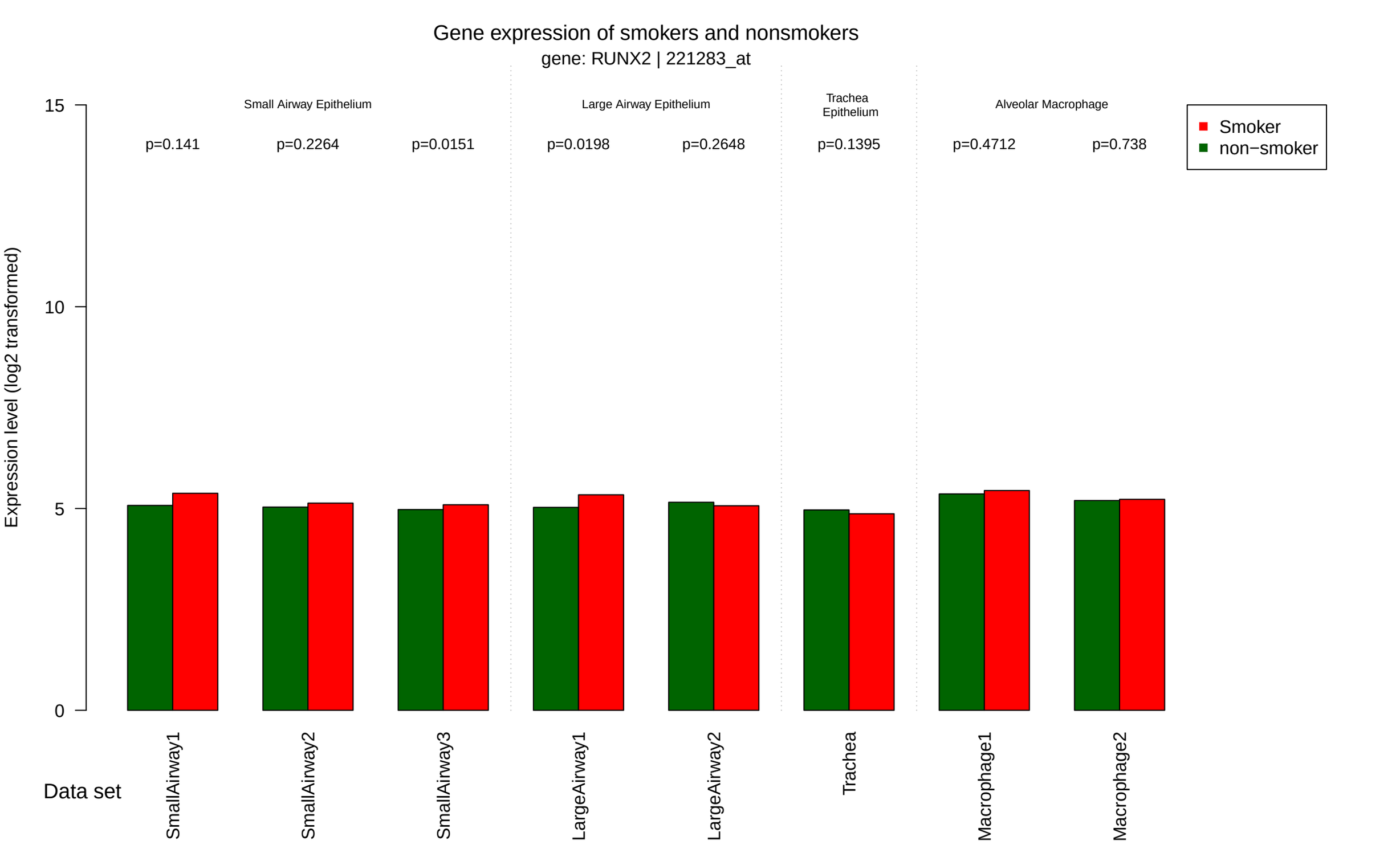
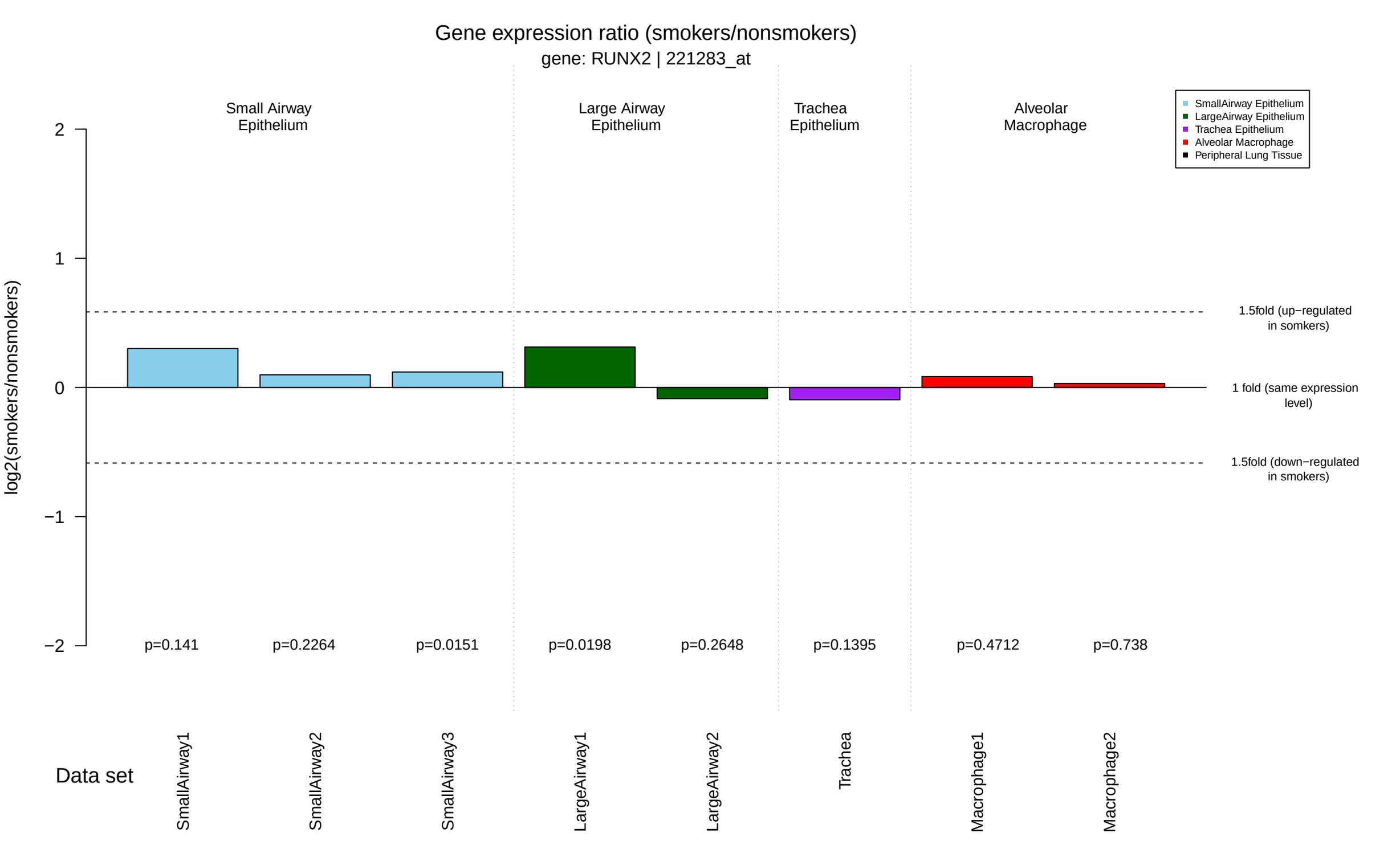
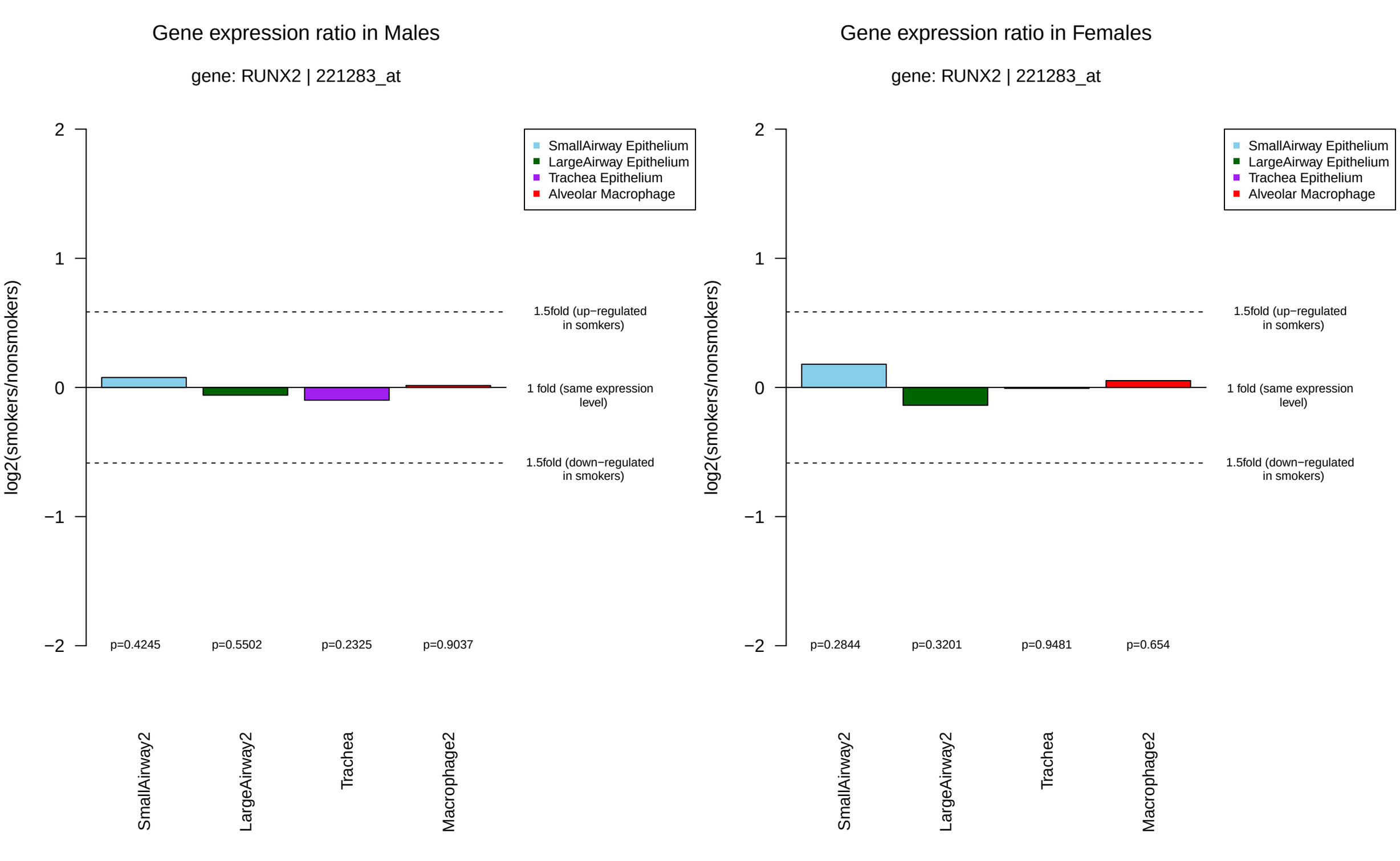
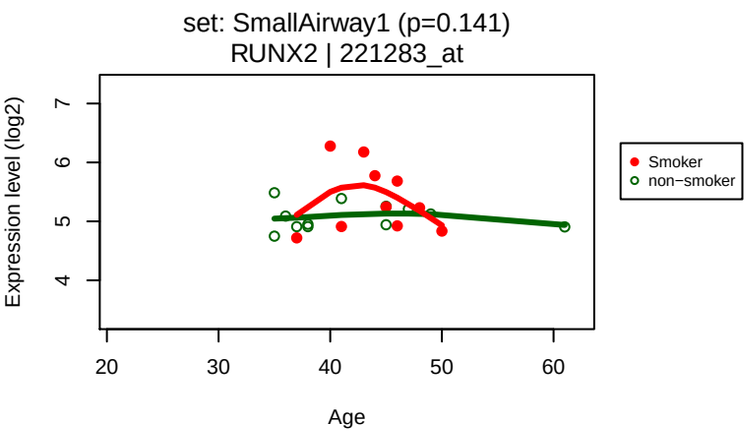
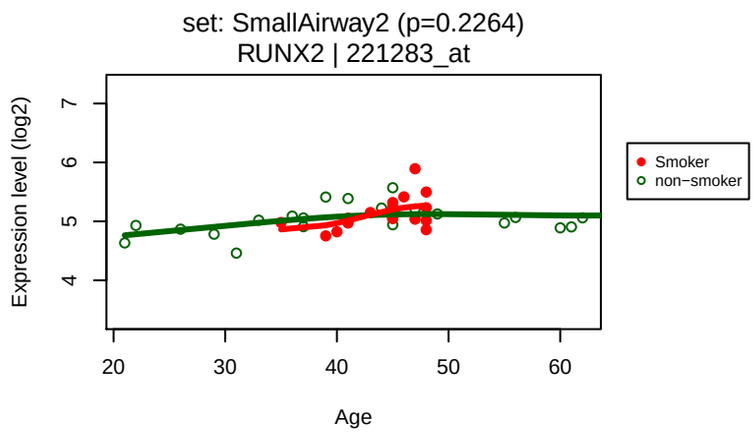
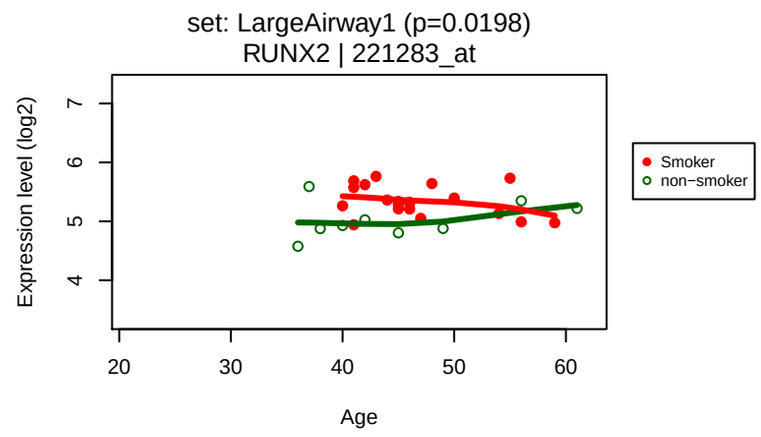

Gene expression of smokers and nonsmokers
gene: RUNX2 | 221283_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.141
p=0.2264
p=0.0151
p=0.0198
p=0.2648
p=0.1395
p=0.4712
p=0.738
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

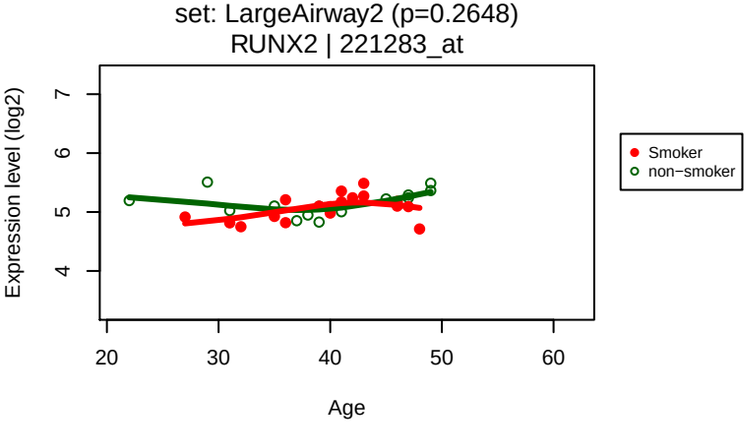

Gene expression ratio (smokers/nonsmokers)
gene: RUNX2 | 221283_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.141
p=0.2264
p=0.0151
p=0.0198
p=0.2648
p=0.1395
p=0.4712
p=0.738
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

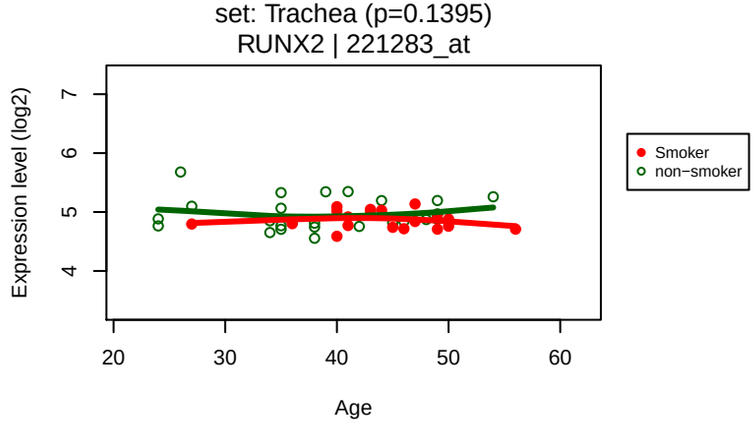

Gene expression ratio in Males
gene: RUNX2 | 221283_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4245
p=0.5502
p=0.2325
p=0.9037
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RUNX2 | 221283_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2844
p=0.3201
p=0.9481
p=0.654
SmallAirway2
LargeAirway2
Trachea
Macrophage2

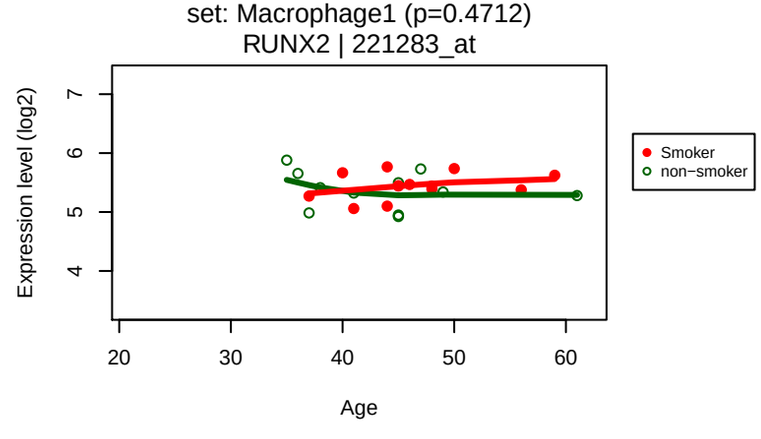

set: SmallAirway1 (p=0.141)
RUNX2 | 221283_at
Expression level (log2)
Age
Smoker
non−smoker

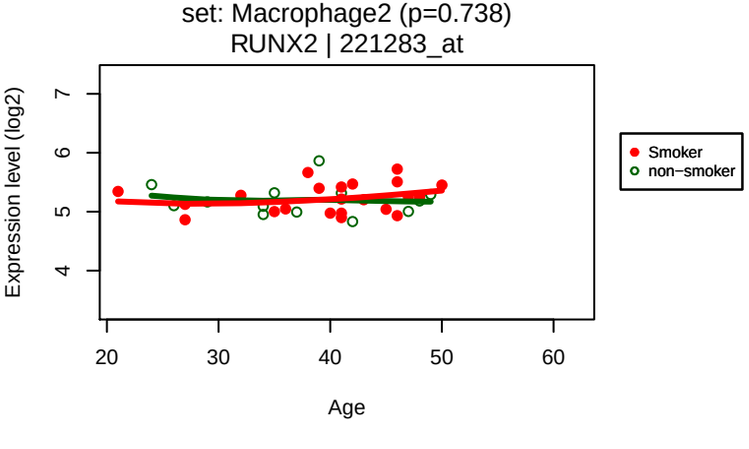

set: SmallAirway2 (p=0.2264)
RUNX2 | 221283_at
Expression level (log2)
Age
Smoker
non−smoker

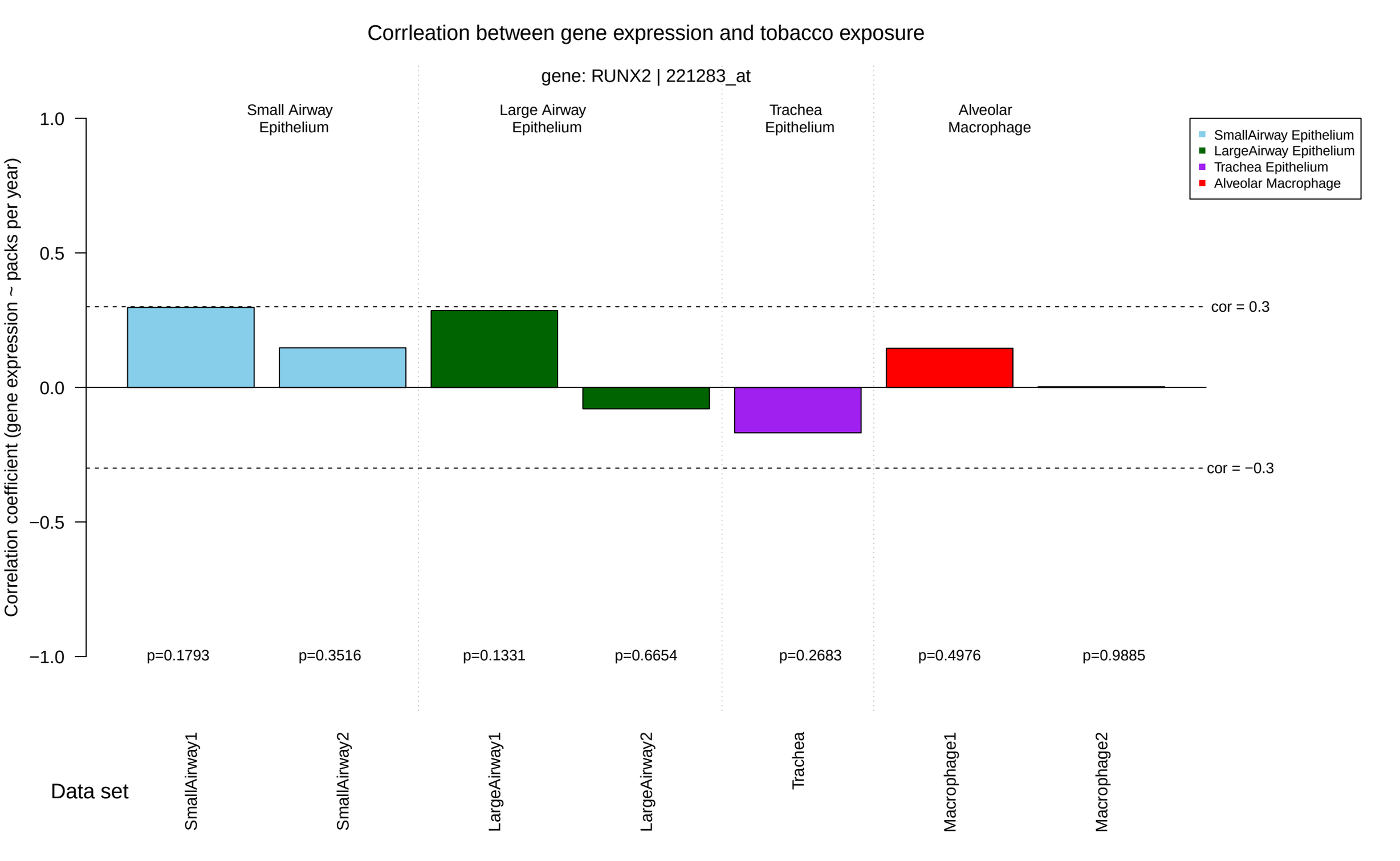

set: LargeAirway1 (p=0.0198)
RUNX2 | 221283_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2648)
RUNX2 | 221283_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1395)
RUNX2 | 221283_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4712)
RUNX2 | 221283_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.738)
RUNX2 | 221283_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RUNX2 | 221283_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1793
p=0.3516
p=0.1331
p=0.6654
p=0.2683
p=0.4976
p=0.9885
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2